Md Ashraful Alam

Dhaka, Bangladesh | ashrafulalam355092@gmail.com | (+880) 1706355092

Portfolio | **LinkedIn** | **GitHub**

Profile

As a Final year Computer Science student at Brac University, I combine a rigorous academic foundation with practical research experience in cutting edge technologies. I am deeply passionate about architecting intelligent systems specializing in Data Science, Machine Learning, and AI while maintaining a versatile technical toolkit that spans Web Development, and Computer Architecture. Now, my focus is on refining my technical expertise and developing strong leadership capabilities to contribute to high impact, forward thinking projects. I thrive at the intersection of innovation and execution.

Skills

Programming Languages: Python, PHP, C
Databases: MySQL
Libraries & Frameworks: TensorFlow, scikit-learn, OpenCV, Flask, NumPy, Pandas, NLTK, Matplotlib
Machine Learning & Data Science: Computer Vision, NLP, Object Detection, ANN, DNN, CNN, RNN, LSTM, GRU, Data Collection, Processing, Analysis, Visualisation
Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms
CSS Frameworks: CSS
Tools & Platforms: GitHub, Hugging Face
Productivity Tools: Google Workspace, MS Office, Canva

Education

BRAC University — 2022 – Sep 2026
BSc in Computer Science — *Dhaka, Bangladesh*
- GPA: 3.03 / 4.00

Govt. Syed Hatem Ali College — 2018 – 2020
Higher Secondary Certificate (HSC) — *Barishal, Bangladesh*
- GPA: 5.00 / 5.00

Shaheed Arju Moni Govt. Secondary School — 2016 – 2018
Secondary School Certificate (SSC) — *Barishal, Bangladesh*
- GPA: 5.00 / 5.00

Research Experience

Undergraduate Researcher — Ongoing
BRAC University, Department of Computer Science and Engineering
- **Title:** Asynchronous Federated Learning-Based Crop Yield Prediction for Sustainable and Energy-Efficient Smart Agriculture
- **Research Objective**
- Asynchronous Aggregation: Eliminate idle time and latency by allowing the global server to update dynamically without waiting for slower, heterogeneous IoT devices.
- Ultra-Low Latency: Minimize communication delays using delta-based updates and a lightweight protocol to keep transmission times below a maximum threshold
- Enhanced Privacy Protection: Secure sensitive farm data through a dual-layer strategy of Differential Privacy and Homomorphic Encryption to prevent inference attacks.
- Improved Energy Consumption: Ensure system sustainability via quantization-based model compression and power-aware client selection to reduce computational load.

Projects

MediCo - Rural Healthcare Platform

- Developed an web platform designed to bridge rural communities with essential healthcare, integrating patient portals, doctor scheduling, and online pharmacy management.
- Built a secure management to support user authentication, interactive symptom checking, and health education resources.
- Tools Used:*Python, Flask, SQL, HTML, CSS*

Hybrid Deep Learning Framework for Economic Time SeriesPrediction

- Engineered and preprocessed macroeconomic time series data from the FRED API, constructing a deep Autoencoder and applying KMeans clustering to compress high-dimensional sequences into latent features.
- Developed a hybrid CNN-LSTM forecasting architecture that integrates clustered contextual data to enhance sequence prediction, achieving a 98.34% accuracy rate in forecasting the Consumer Price Index (CPI).
- Tools Used:*Python, PyTorch, scikit-learn, pandas, NumPy, matplotlib, FRED API*

A Machine Learning Pipeline for Loan Approval Prediction

- Engineered an end-to-end machine learning pipeline to predict loan approval status, implementing comprehensive data preprocessing, feature scaling, and stratified splitting to handle imbalanced class distributions.
- Conducted a comparative evaluation of multiple classification algorithms to optimize predictive performance, selecting the ideal model based on superior overall accuracy and AUC scores.
- Tools Used:*Python,Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn*

Hotel Management System

- Developed a web application to efficiently manage hotel room inventory, track guest information, and streamline the complete booking lifecycle.
- Implemented secure, role-based administrative access controls for CRUD operations on reservations and utilized a custom HTML/CSS frontend to deliver a user-friendly interface.
- Tools Used: *PHP, HTML, CSS, MySQL*

Extracurricular

Finance Department | *BRAC University Computer Club*

- Contributed to the Finance Department of BRAC University Computer Club, supporting budget planning and financial coordination for club events and initiatives.
- Collaborated with cross-functional club panels (technical, event management, and outreach) to align financial planning with organizational goals.

Certifications and Achievement

Associate Data Scientist | *DataCamp*
AWS Academy Cloud Foundations | *AWS Academy*

Language Skills

Bangla: Native (Mother Tongue)
English: Independent User (Listening: B2 | Reading: B2 | Writing: B1 | Speaking: A2)
Spanish: Basic User